Ai For Detection

⚙️ 1. AI DECISION SYSTEM (CORE IDEA)

```
Final Decision = Confidence Score (0–100)

If score > threshold → Emergency = TRUE
Else → FALSE
```

🧩 2. INPUT SIGNALS (WHAT AI USES)

You already collect:

🎤 Audio

📷 Video

📳 Motion

Now extract **simple features** from each.

🔊 3. AUDIO ANALYSIS (EASIEST + HIGH IMPACT)

What to detect:

✅ Keywords:

- "help"
- "bachao"
- "danger"

- "please"

👉 Use:

- Pre-trained Speech-to-Text (Google API / Whisper)

✅ Panic Detection (Simple Version)

Instead of complex ML:

Check:

- High volume
- Sudden spikes
- Fast speech

Output:

```
keyword_detected → +40
high_volume → +20
panic_pattern → +20
```

📳 4. MOTION ANALYSIS (VERY IMPORTANT)

From accelerometer:

Detect:

- Sudden jerks
- Continuous shaking
- Fall detection (optional)

Output:

```
strong_shake → +30
multiple movements → +20
```

🎥 5. VIDEO ANALYSIS (KEEP IT SIMPLE)

⚠️ Don’t try face emotion detection (too complex)

Instead:

Detect:

- Rapid movement
- Camera instability
- Low light + movement

Output:

```
high_motion → +20
unstable frames → +20
```

🧮 6. FINAL DECISION ENGINE

Combine everything:

```
score = audio + motion + video

IF score >= 60 → Emergency TRUE
ELSE → FALSE
```

🔁 7. COMPLETE AI FLOW

```
1. Receive audio/video/motion
2. Extract features
3. Assign scores
4. Sum total
5. Compare with threshold
6. Return decision
```

💻 8. HOW YOU ACTUALLY BUILD THIS (REAL STACK)

🎤 Audio Processing

Use:

- Google Speech-to-Text API
  OR
- OpenAI Whisper (better)

Then:

- Check keywords in text

📳 Motion

- Already available in app (no AI needed)
- Just threshold logic

🎥 Video

Simplest approach:

Use:

- OpenCV

Check:

- Frame difference
- Movement intensity

🔧 9. BACKEND AI SERVICE (STRUCTURE)

```
POST /analyze-emergency

Input:
{
  audio,
  video,
  motion_data
}

Output:
{
  is_emergency: true,
  confidence_score: 75
}
```

🔥 10. HACKATHON MVP (WHAT YOU SHOULD ACTUALLY BUILD)

👉 Keep it SIMPLE:

MUST:

- Keyword detection from audio ✅
- Motion detection ✅

OPTIONAL:

- Basic video motion detection

🎯 BEST SHORTCUT (SECRET WEAPON)

If you are short on time:

👉 Skip video AI completely

Just use:

```
Keyword detected → TRUE
OR
Strong shake → TRUE
```

Then say:

“We designed modular AI system where more signals can be plugged in”

👉 Judges will LOVE this

🧠 WHAT YOU SAY TO JUDGES

👉

“Instead of relying on a single unreliable AI model, we built a multi-signal confidence-based system combining audio keywords, motion intensity, and contextual cues to ensure accurate emergency detection with minimal false positives.”

⚡ REALITY CHECK (VERY IMPORTANT)

Even big companies struggle with:

- Emotion detection
- Real danger classification

So don’t try to:

- Build complex deep learning ❌

- Train datasets ❌

👉 Build **smart logic + small AI usage** ✅